Team name	[Your Team Name]
Project title	NookSync: Live Study Space and Peer Study Pod Matcher
Selected case	Case 1: Smarter Campus and Learning
Members	[Member A (leader), Member B, Member C, Member D]

Abstract

1. Problem. Students frequently waste valuable time between lectures searching for available group study rooms or quiet collaborative tables across campus buildings, only to find rooms either unrecorded in any central system or reserved by "ghost bookings" where no one shows up. At the same time, students who want to review difficult courses often study alone because they cannot easily see which peers taking the same subject have matching free hours on campus.

2. Intended Users. Undergraduate students who have 1-to-3-hour gaps between lectures during peak campus hours (10:00 to 17:00) and need a space or study partners for coursework, alongside building administrators who manage shared discussion rooms.

3. Supporting Observations or Evidence. Based on direct campus observation across [N, e.g., 3] shared study areas and a survey of [M, e.g., 38] students [1], [X, e.g., 30] respondents reported spending 15 to 30 minutes walking between floors to find an open table with power outlets. We also observed that existing room reservations are handled through separate spreadsheets or messaging admins per building, with an estimated 25% of booked slots left unoccupied due to no-shows.

4. Proposed Solution and Core User Flow. NookSync is a mobile-first web application that combines live campus room availability, QR check-in verification, and course-based study pod matching. The core user flow has four steps: (1) A student logs in with their campus account and uploads their semester course schedule PDF once to detect their weekly free time slots and enrolled courses. (2) On the interactive building floorplan, the student views real-time room statuses: Green (Available), Yellow (Open Study Pod accepting peers for a specific course), or Red (Full). (3) The student either books an available room for up to two hours or joins an existing Open Study Pod studying the same course during their free gap. (4) Upon arriving, at least one member must scan a dynamic QR code at the room within 15 minutes; otherwise, the system automatically releases the room for other students. The initial version covers two campus buildings and 15 study spaces.

5. How It Differs from Existing Alternatives. Current campus booking forms operate in isolated silos without real-time occupancy updates or no-show release rules, while group chats require manual coordination to find study partners. NookSync uniquely links physical space booking with automated 15-minute QR check-in enforcement and schedule-aware "Open Study Pod" matching.

6. Planned Technologies. A mobile-responsive React and Next.js web application using Supabase PostgreSQL and Supabase Realtime WebSockets for instant status updates, interactive floorplans built with the tldraw canvas SDK, and a Shadcn UI dashboard for building staff. We plan to use MarkItDown and Google LangExtract (powered by Gemini) solely during onboarding to parse course codes and free time blocks from student schedule PDFs.

7. Expected Benefit and Development Plan. Expected benefit: cut average room and study-partner search time to under 3 minutes and reduce unclaimed ghost bookings by over 60%, measured from check-in logs during a two-week pilot with 25 students. Plan: Week 1 database, floorplan UI, and booking logic; Week 2 QR auto-release and schedule parser; Week 3 pilot test and bug fixes. The main risk is physical QR code tampering, which we mitigate using rotating time-based tokens displayed on a staff tablet or verified via campus Wi-Fi/geolocation bounds.

Source Acknowledgments. [1] Campus study space observation and student survey conducted by the team (November 2026).